

Inventive Media

Build Responsive Websites with Bootstrap

Day 3

Key Topics

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Who is bootstrap?

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What is Bootstrap?

What is Bootstrap?

Bootstrap is a free and open-source CSS framework used for developing responsive, mobile-first websites and web applications.

It provides a collection of pre-designed HTML, CSS, and JavaScript components, making it easier and faster for developers to build consistent and visually appealing interfaces.

The Competitors



Tailwind CSS
Utility first class



SASS CSS
Super CSS



Bulma CSS
CSS Framework



Materialize CSS
Google

The Bootstrap Setup

Ways to setup bootstrap In your project



Via CDN

A quick way to include Bootstrap's CSS and JS files in your project by linking to them online. Instead of downloading the files locally

Via Download files

Manually download the Bootstrap CSS and JS files from the official website and include them in your project folder.

Via installation (NPM)

Installing Bootstrap via npm lets you manage it as a project dependency, pin versions, and integrate with your build tool (Webpack, Rollup, Vite, etc.).

Install Bootstrap via CDN

CDN via jsDelivr

Skip the download with [jsDelivr](#) to deliver cached version of Bootstrap's compiled CSS and JS to your project.

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.7/dist/css/bootstrap.min.css" rel="style"
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.7/dist/js/bootstrap.bundle.min.js" integrity="
```

If you're using our compiled JavaScript and prefer to include Popper separately, add Popper before our JS, via a CDN preferably.

```
<script src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.11.8/dist/umd/popper.min.js" integrity="
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.7/dist/js/bootstrap.min.js" integrity="
```


Install Bootstrap via Download files

Compiled CSS and JS

Download ready-to-use compiled code for **Bootstrap v5.3.7** to easily drop into your project, which includes:

- Compiled and minified CSS bundles (see [CSS files comparison](#))
- Compiled and minified JavaScript plugins (see [JS files comparison](#))

This doesn't include documentation, source files, or any optional JavaScript dependencies like Popper.

Download



Download files here

Install Bootstrap via NPM

Package managers

Pull in Bootstrap's **source files** into nearly any project with some of the most popular package managers. No matter the package manager, Bootstrap will **require a [Sass compiler](#) and [Autoprefixer](#)** for a setup that matches our official compiled versions.

npm

Install Bootstrap in your Node.js powered apps with [the npm package](#):

```
$ npm install bootstrap@5.3.7
```



The Bootstrap Syntax

How to use?

Normally you can add a predefined class names in your HTML element to style and layout your content without writing a custom CSS.

Because these classes are styled already for you to rapidly style your components with ease and consistently reuse the classes when you needed.



```
<button class="btn btn-primary">Primary Button</button>
```

Typography

Bootstrap provides a wide range of utility classes that allow you to style text efficiently — including formatting, alignment, sizing, color, and text transformation — all without the need for custom CSS. These predefined classes make it easy to implement consistent and responsive typography across your project.

```
<p class="h1 fw-bold text-center">I'm heading tag</p>
```

Typography

Class name	Description
.h1 to .h6	Applies heading styles (<h1> to <h6>) to any element.
.display-1 to .display-6	Larger, bolder headings for showcasing headlines
.lead	Styles paragraph text to stand out, slightly larger font and lighter weight
.small	Applies the styles <small> tag
.mark	Applies a highlighted text
.fw-bold (font-weight)	Applies bold font weight
.fw-normal (font-weight)	Applies normal font weight

Typography

Class name	Description
<code>.fw-light</code> (font-weight)	Applies a light font weight
<code>.fst-italic</code> (font-style)	Applies an italic style text
<code>.text-decoration-underline</code>	Applies underline text
<code>.text-decoration-line-through</code>	Applies line through text
<code>.text-center</code>	Centers a text
<code>.text-start</code>	Align text to left (start of container)
<code>.text-end</code>	Align text to right (end of container)

Typography

Class name	Description
.lh-1 (line-height)	Adjust a line height tight
.lh-sm (line-height)	Adjust a line height small
.lh-base (line-height)	Adjust a line height normal
.lh-lg (line-height)	Adjust a line height large

The Utility Classes

Colors

Bootstrap provides a set of predefined colors and classes to style text and background elements. These include theme colors like primary, secondary, success, danger, etc.

```
<p class="text-primary fw-bold">  
  I'm text Primary  
</p>
```

I'm text Primary

Colors

Class name	Description
.text-primary	Applies the primary theme color to text (default: blue)
.text-secondary	Applies a secondary text color (default: gray)
.text-danger	Red color used for errors, warnings, or destructive actions
.text-success	Green color used for success messages or positive actions
.text-warning	Yellow/orange color for caution or alerts
.text-info	Light blue for informational messages
.text-muted	Muted gray text, for less emphasis

Background colors

Class name	Description
.bg-primary	Applies primary background color
.bg-secondary	Applies secondary background color
.bg-danger	Red background, for errors or alerts
.bg-success	Green background, for positive messages
.bg-warning	Yellow/orange background, for warnings
.bg-info	Light blue background, for info sections
.bg-transparent	Makes background transparent

Spacing

Bootstrap includes a wide range of shorthand responsive margin, padding, and gap utility classes to modify an element's appearance.

Bootstrap includes a 5 level sets of spacing to all sides that you can use to customize any space in your element.

[See documentations](#)

```
<p class="p-5 m-5 fw-bold">  
  I'm text with space  
</p>
```



Padding

Class name	Description
.p	Applies padding on all sides
.pt (padding-top)	Applies padding on top
.pe (padding-right / end)	Applies padding on right
.pb (padding-bottom)	Applies padding on bottom
.ps (padding-left / start)	Applies padding on left
.px (horizontal)	Applies padding (left / right) horizontal
.py (vertical)	Applies padding (top / bottom) vertical

Margin

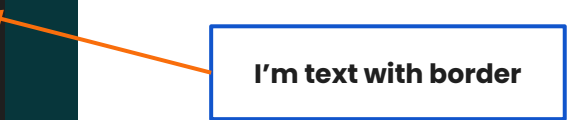
Class name	Description
.m	Applies margin on all sides
.mt (padding-top)	Applies margin on top
.me (padding-right / end)	Applies margin on right
.mb (padding-bottom)	Applies margin on bottom
.ms (padding-left / start)	Applies margin on left
.mx (horizontal)	Applies margin (left / right) horizontal
.my (vertical)	Applies margin (top / bottom) vertical

Borders

Use border utilities to add or remove an element's borders. Choose from all borders or one at a time.

See documentations

```
<p class="border border-2 border-primary" >  
  I'm text with border  
</p>
```



I'm text with border

Border

Class name	Description
<code>.border</code>	Applies border on all sides
<code>.border-0</code>	Removes border on all sides
<code>.border-top</code>	Applies border on top
<code>.border-end (right)</code>	Applies border on right
<code>.border-bottom</code>	Applies border on bottom
<code>.border-start (left)</code>	Applies border on left
<code>.border-5</code>	Applies border level of thickness (1-5)

Border

Class name	Description
<code>.border-primary</code>	Applies a color to border (bootstrap colors)
<code>.rounded-0</code>	Removes border radius
<code>.rounded-1</code>	Applies rounded sizes (1-3 scale)
<code>.rounded-circle</code>	Makes full circle
<code>.rounded-top</code>	Applies rounded corners on top
<code>.rounded-bottom</code>	Applies rounded corners on bottom
<code>.rounded-end</code>	Rounds right/end corners only
<code>.rounded-start</code>	Rounds left/start corners only

Box shadows

Used to add shadow effects around an element's frame. Bootstrap provides utility classes to easily apply shadows of varying sizes and intensities to elements.

See documentations

```
<p class="p-3 shadow">  
  I'm medium box shadow  
</p>
```

I'm medium box shadow

Box shadows

Class name	Description
.border-primary	Applies a color to border (bootstrap colors)
.rounded-0	Removes border radius
.rounded-1	Applies rounded sizes (1-3 scale)
.rounded-circle	Makes full circle
.rounded-top	Applies rounded corners on top
.rounded-bottom	Applies rounded corners on bottom
.rounded-end	Rounds right/end corners only
.rounded-start	Rounds left/start corners only

Container class

Container

Containers are a fundamental building block of Bootstrap that contain, padding, and align your content within a given device or viewport.

[See documentations](#)

```
<p class="container px-3">  
  I'm a small container  
</p>
```

I'm small container

Container

	Extra small <576px	Small ≥576px	Medium ≥768px	Large ≥992px	X-Large ≥1200px	XX-Large ≥1400px
<code>.container</code>	100%	540px	720px	960px	1140px	1320px
<code>.container-sm</code>	100%	540px	720px	960px	1140px	1320px
<code>.container-md</code>	100%	100%	720px	960px	1140px	1320px
<code>.container-lg</code>	100%	100%	100%	960px	1140px	1320px
<code>.container-xl</code>	100%	100%	100%	100%	1140px	1320px
<code>.container-xxl</code>	100%	100%	100%	100%	100%	1320px
<code>.container-fluid</code>	100%	100%	100%	100%	100%	100%

Container

Class name	Description
.container	Full width until $\geq 576\text{px}$, then fixed-width
.container-md	Full width until $\geq 768\text{px}$, then fixed-width
.rounded-1	Applies rounded sizes (1-3 scale)
.container-lg	Full width until $\geq 992\text{px}$, then fixed-width
.container-xl	Full width until $\geq 1200\text{px}$, then fixed-width

Grid layout

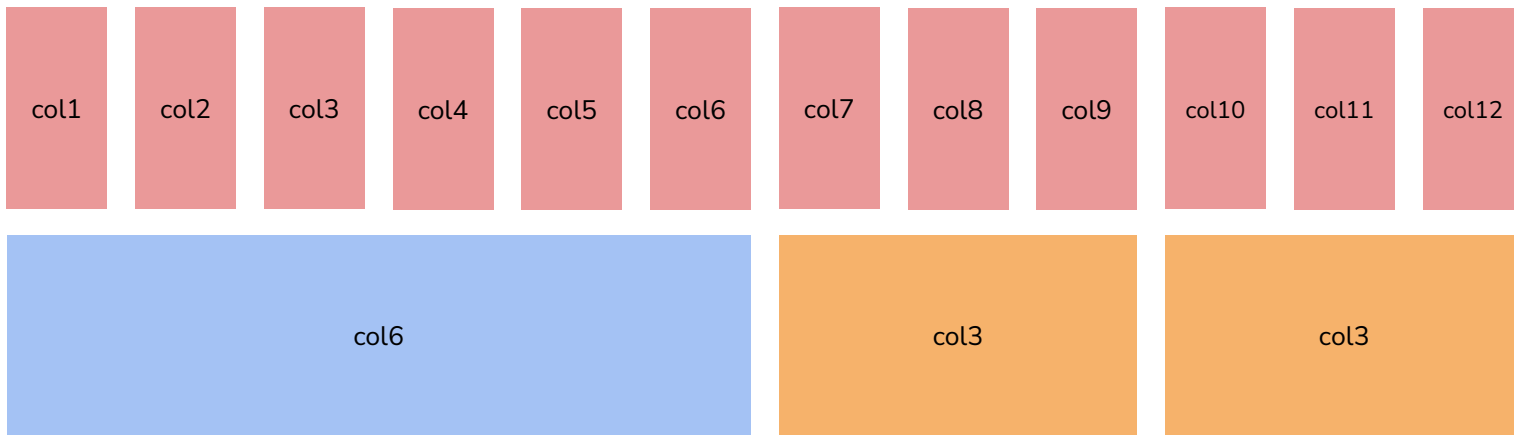
Grid layout

Bootstrap's grid system is flexbox-based and divides the page layout into 12 equal columns. You can combine these columns to create responsive layouts.



Grid layout

You can create any combination of columns as long as it will fit the 12 column grid standard in bootstrap.



Grid layout

Class name	Description
.container	Centers content with fixed width at each breakpoint
.container-fluid	Full-width container, spans 100% of the viewport
.row	Defines a horizontal group of columns (clears floats, adds negative margins)
.col	Auto-sized column that shares available space equally with others
.col-1 to .col-12	Defines column width (e.g., .col-6 = 6/12 or 50% width)
.g-0 to .g-5	Gutter spacing between columns (g stands for gutter)
.gx-0 to .gx-5	Horizontal gutter spacing only

Grid layout

Class name	Description
<code>.gy-0</code> to <code>.gy-5</code>	Vertical gutter spacing only
<code>.align-items-start</code>	Aligns column contents to the top vertically
<code>.align-items-center</code>	Vertically centers column contents
<code>.align-items-end</code>	Aligns column contents to the bottom
<code>.justify-content-start</code>	Aligns columns to the left horizontally
<code>.justify-content-center</code>	Centers columns horizontally
<code>.justify-content-end</code>	Aligns columns to the right horizontally

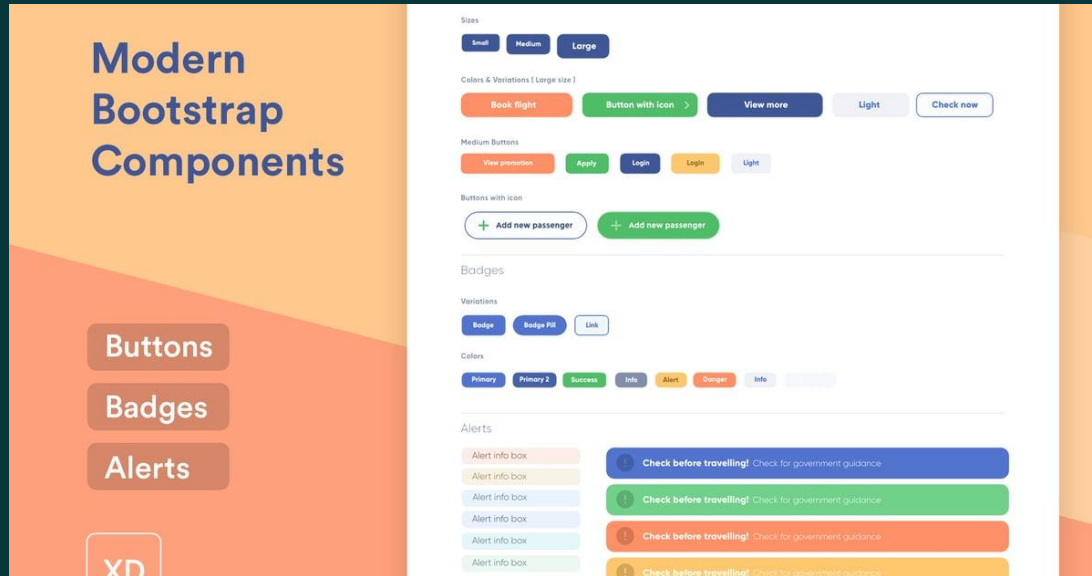
Grid layout

Class name	Description
<code>.justify-content-between</code>	Evenly distributes columns with space between
<code>.justify-content-around</code>	Even spacing around columns
<code>.justify-content-evenly</code>	Equal spacing between columns

Components

Components

It is a pre-styled, reusable user interface (UI) element or building block. These components are designed to provide a consistent and responsive design framework, facilitating the rapid development of web applications. They are built using HTML, CSS, and optionally JavaScript, and come with predefined styles and behaviors.



Components

Class name	Description
Navbar	Responsive navigation header with links, branding, togglers.
Accordion	Full-width container, spans 100% of the viewport
Alerts	Provides contextual feedback messages (e.g., success, warning, danger).
Badges	Small count or labeling element, often used for notifications.
Breadcrumb	Navigation bar that indicates the current page's location in a hierarchy.
Buttons	Standard clickable buttons with customizable styles and sizes.
Button Group	Groups a series of buttons together on a single line.

Components

Class name	Description
Card	Flexible content container with options for headers, footers, and images.
Accordion	Full-width container, spans 100% of the viewport
Carousel	A slideshow component for cycling through elements like images or text.
Collapse	Expands/collapses content (often used in accordions or toggles).
Dropdowns	Toggleable menu for showing a list of links or actions.
Forms	Custom form controls like inputs, checkboxes, selects, etc.
Input Group	Combine form inputs and add-ons (e.g., icons, labels) into one line.

Components

Class name	Description
List Group	Flexible component for displaying lists of elements.
Modal	Popup/dialog box that overlays the current page.
Offcanvas	Sidebar-like panel that appears from the side of the screen.
Pagination	Provides navigation links for paginated content.
Placeholders	Lightweight placeholder UI for loading states.
Popovers	Small overlay with more content triggered on click or hover.

Components

Class name	Description
Progress	Shows progress bars with optional labels or animations.
Scrollspy	Highlights nav items based on scroll position.
Spinners	Loading indicators for asynchronous content.
Toasts	Lightweight notifications that slide in and disappear automatically.
Tooltips	Small text boxes that appear on hover/focus to describe elements.

Available resources

Here are available resources that can help you to build components, pages, layout in bootstrap

See documentations

Bootstrap generators

- Bootstrap cheatsheet
- LayoutIt!

Bootstrap templates

- Start Bootstrap



Thank you!